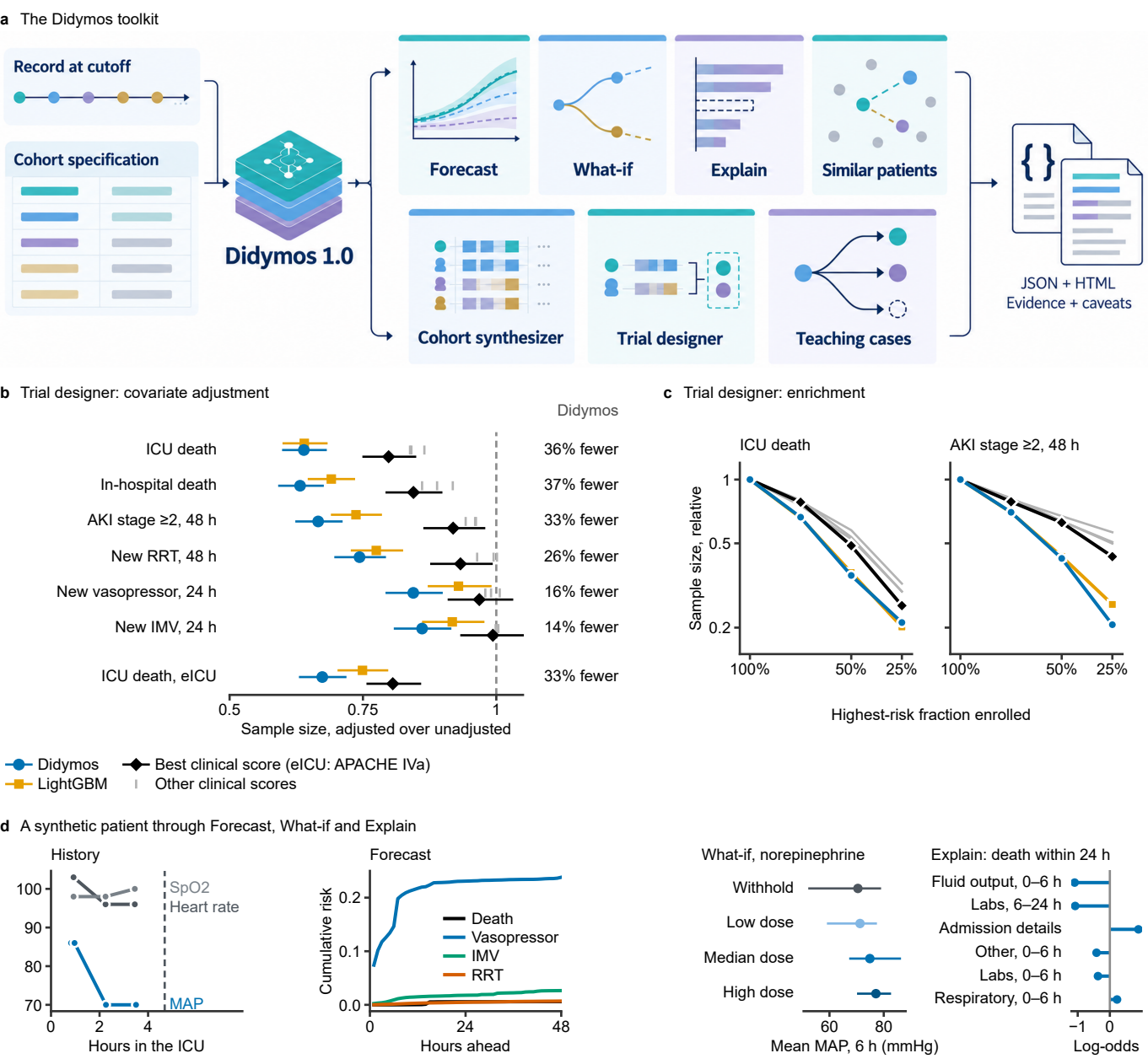

a The Didymos toolkit
Record at cutoff
Cohort specification
Didymos 1.0
Forecast
What-if
Explain
Similar patients
Cohort synthesizer
Trial designer
Teaching cases
JSON + HTML
Evidence + caveats
b Trial designer: covariate adjustment
Didymos
ICU death
In-hospital death
AKI stage ≥2, 48 h
New RRT, 48 h
New vasopressor, 24 h
New IMV, 24 h
ICU death, eICU
36% fewer
37% fewer
33% fewer
26% fewer
16% fewer
14% fewer
33% fewer
0.5
0.75
1
Sample size, adjusted over unadjusted
Didymos
LightGBM
Best clinical score (eICU: APACHE IVa)
Other clinical scores
c Trial designer: enrichment
ICU death
AKI stage ≥2, 48 h
Sample size, relative
1
0.5
0.2
100%
50%
25%
Highest-risk fraction enrolled
d A synthetic patient through Forecast, What-if and Explain
History
SpO2
Heart rate
MAP
100
90
80
70
0
2
4
Hours in the ICU
Forecast
Cumulative risk
0.2
0.1
0.0
Death
Vasopressor
IMV
RRT
0
24
48
Hours ahead
What-if, norepinephrine
Withhold
Low dose
Median dose
High dose
60
80
Mean MAP, 6 h (mmHg)
Explain: death within 24 h
Fluid output, 0–6 h
Labs, 6–24 h
Admission details
Other, 0–6 h
Labs, 0–6 h
Respiratory, 0–6 h
−1
0
Log-odds